

20 DNA Tools and Biotechnology

KEY CONCEPTS

- 20.1** DNA sequencing and DNA cloning are valuable tools for genetic engineering and biological inquiry *p. 416*
- 20.2** Biologists use DNA technology to study gene expression and function *p. 423*
- 20.3** Cloned organisms and stem cells are useful for basic research and other applications *p. 428*
- 20.4** The practical applications of DNA-based biotechnology affect our lives in many ways *p. 433*

Study Tip

Apply what you've learned: Write down a gene-related question you've wondered about. Then make a table listing techniques you could use to investigate your question and how you would apply them. Below is an example.

Is autism caused by genes, the environment, or both?	
Technique	Application to question
DNA sequencing	-Compare DNA sequences from people with autism with sequences from nonaffected people.

Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 20
- HHMI Video: Genetic Engineering
- Animation: Creating Recombinant DNA
- HHMI Video: Polymerase Chain Reaction (PCR)

For Instructors to Assign (in Item Library)

- Tutorial: Restriction Enzymes, Recombinant DNA, and Gene Cloning
- Activity: Cloning a Gene in Bacteria

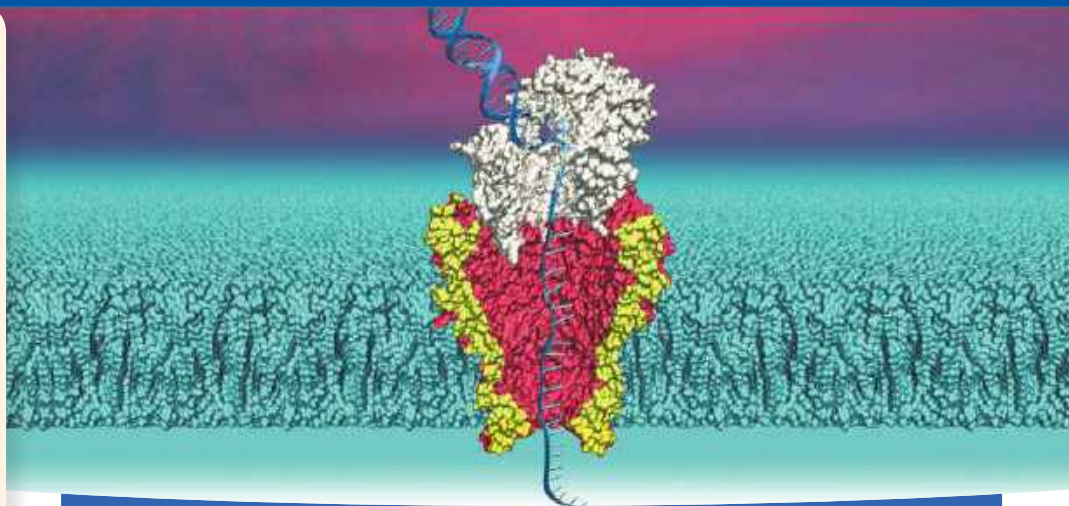
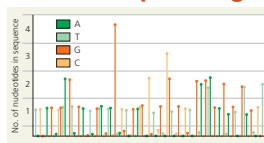


Figure 20.1 This model shows a technique in which a DNA strand is passed through a small pore in a membrane. The resulting changes in an electrical current are used to determine the nucleotide sequence. The first human genome sequence, completed in 2003, took 13 years and cost \$1 billion; the time and cost of genome sequencing have been greatly reduced by improved methods like the one shown here.

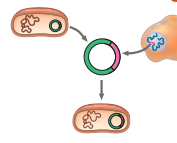
What are the main techniques and applications of biotechnology?

TECHNIQUES

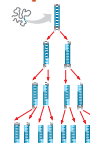
DNA sequencing



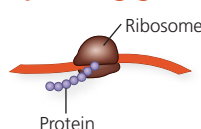
Gene cloning



PCR amplification



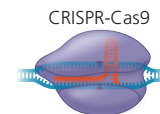
Expressing genes



Analyzing gene expression



Gene editing



APPLICATIONS

Agriculture



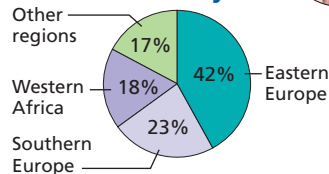
Medicine



Environmental cleanup



Ancestry



Forensics

Victim	Suspect 1	Suspect 2	Crime scene
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—